

# Princeton Liu

510-335-9673 | [princetonliu@g.ucla.edu](mailto:princetonliu@g.ucla.edu) | [linkedin.com/in/princeton-liu/](https://www.linkedin.com/in/princeton-liu/) | [princetonliu.github.io](https://princetonliu.github.io) | [github.com/princetonliu](https://github.com/princetonliu)

## EDUCATION

University of California, Los Angeles

August 2023 – June 2027

*Computer Science B.S & Applied Mathematics B.S: GPA: 3.85/4.00*

*Los Angeles, CA*

- **Coursework:** Machine Learning, Algorithms, Compilers, Probability, Statistics, Numerical Methods

## EXPERIENCE

Research Assistant (Quantum Programming Languages)

January 2026 – Present

*Jens Palsberg Quantum Group (UCLA)*

*Los Angeles, CA*

- Implementing Shor's algorithm in **Strawberry Fields** and **HML** to evaluate simulator **accuracy** and **runtime** characteristics across quantum programming frameworks.

Computer Vision Research Assistant | *PyTorch, numPy*

September 2024 – September 2025

*Computational Vision and Learning Lab (UCLA CVL)*

*Los Angeles, CA*

- Designed and ran experiments using function vectors (via **average indirect effect** and **activation aggregation**) to improve **zero-shot relational reasoning** in CLIP ViT-B/16 by **34% without fine-tuning**.
- Tools/Topics: PyTorch, NumPy, Matplotlib, Vision Language Models, Transformers, Contrastive Learning

Reader (Machine Learning, Programming Languages)

September 2024 – Present

- Graded/gave feedback to **300+** students on data types, control flow, parsing, edge cases, and code quality
- Graded **ML** assignments/exams, focusing on model correctness, implementation, and statistical foundation

Learning Assistant (Data Science)

September 2024 – December 2024

- Taught **80+** undergraduates data science methods and assisted with projects using Sklearn/PyTorch/Pandas

## PROJECTS

Inference & Visualization System for CLIP | *PyTorch, Golang, Docker, AWS EC2*

January 2026 – Present

- Built a **web-based system** for running CLIP ViT-B/16 image-text inference on uploaded user inputs.
- Implemented a backend inference service and **APIs** to execute CLIP and return **outputs** and **attention maps**.
- Containerized environment using Docker and **deployed** on AWS EC2. Public repository available [here].

MiniJava Compiler | *Java 8, Java Tree Builder*

March 2025 – June 2025

- Built end-to-end compiler pipeline: parser, type-checker, Sparrow IR translator, register allocator (linear scan), and RISC-V code generator. Used **OOP** (Java 8) and **DSA** (trees, maps, topological sort, etc.)
- Implemented **static analysis checks** to detect semantic errors, undefined behavior, and invalid memory access

Neural Signal Speech Decoder | *PyTorch, GRUs, Attention*

September 2025 – December 2025

- Designed and evaluated **GRU**-based neural decoders with residual attention, achieving **25%** error reduction on neural signal decoding task. Manuscript, results, and project code available [here].
- Conducted ablations and performance analysis on attention mechanisms and sequence modeling choices.

Brewin Language Series (Brewin, Brewin++, Brewin#) | *Python 3*

September 2024 – December 2024

- Built end-to-end **interpreters** supporting recursion, type systems, lazy evaluation, caching, and error handling
- Gained experience in control flow and error detection, applicable to static/fuzzing tool development.

## AWARDS AND HONORS

United States of America Mathematics Olympiad (USAMO) Qualifier

2022

*Top 250 scorers in AMC-12/AIME out of 300,000+ highschool students*

## TECHNICAL SKILLS

**Languages:** C++20, Python 3, Java 8, SQL, C, GoLang, JavaScript, MATLAB, Haskell, Prolog, R

**Libraries/Frameworks:** PyTorch, net/http, FastAPI, NumPy, Docker, Nginx, Matplotlib, Scikit-learn, Pandas

## HOBBIES

Board games, Minesweeper (Top 5000 Winter '26 Season), Boulderling, Poker, Skateboarding!